GVIM Chemical Intelligence Agent

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Chemical Intelligence Agent

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Project Background

ChatGPT has ushered in the era of large AI models.

- OpenAI company has developed a large-scale language model:
  – It learns language patterns from a massive corpus, enabling it to generate human-like text.
- It functions as a conversational AI capable of understanding natural language and engaging in high-quality multi-turn dialogues.
- –Its human-like responses are astonishing, making it a milestone breakthrough in artificial intelligence.

Bill Gates: The Importance of ChatGPT is Comparable to the Invention of the Internet

Released on November 30, 2022, it gained 1 million users in just 5 days. By January of this year, its monthly active users had reached 100 million.

Limitations of ChatGPT

- Unable to incorporate new knowledge in real time
- Lacks expertise in highly specialized fields
- Reasoning and computational abilities are still insufficient, leading to possible nonsensical responses
- Only supports text generation
- High computational cost
- ……

However, most of these issues are engineering challenges and do not present insurmountable theoretical obstacles.

Project Background

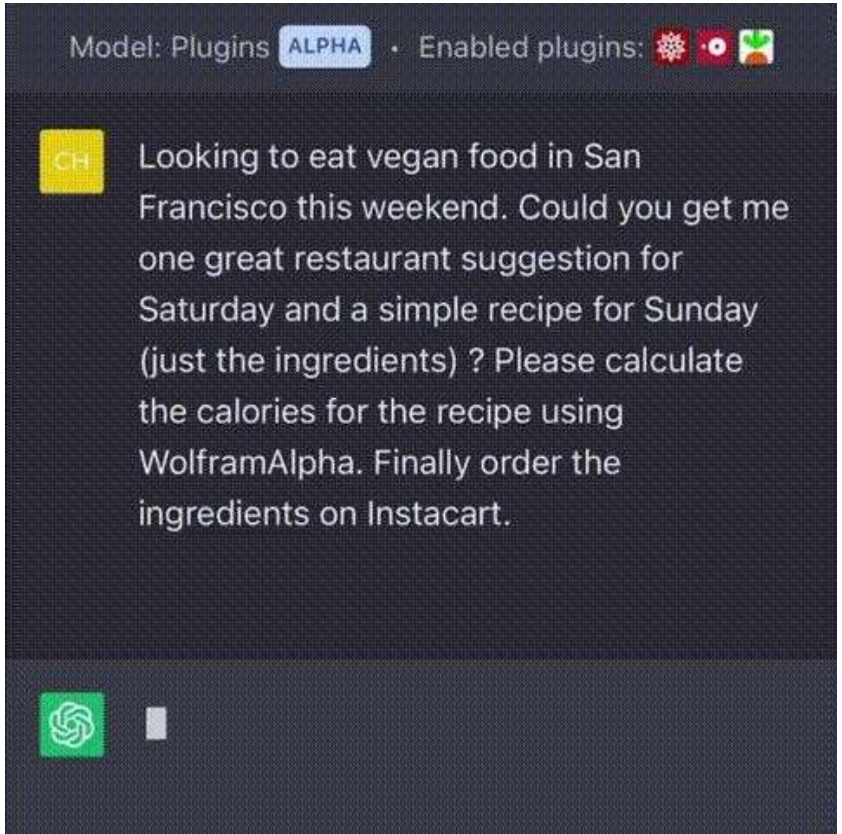
Tools Use

To address the **limitations** of large models mentioned above, we can reference how humans effectively handle complex and subtle pattern recognition tasks. Humans can achieve this by supplementing the prior knowledge in their minds with tools like **books, search engines, or calculators**, and only then formulating a conclusion (for example, "The image describes XX").

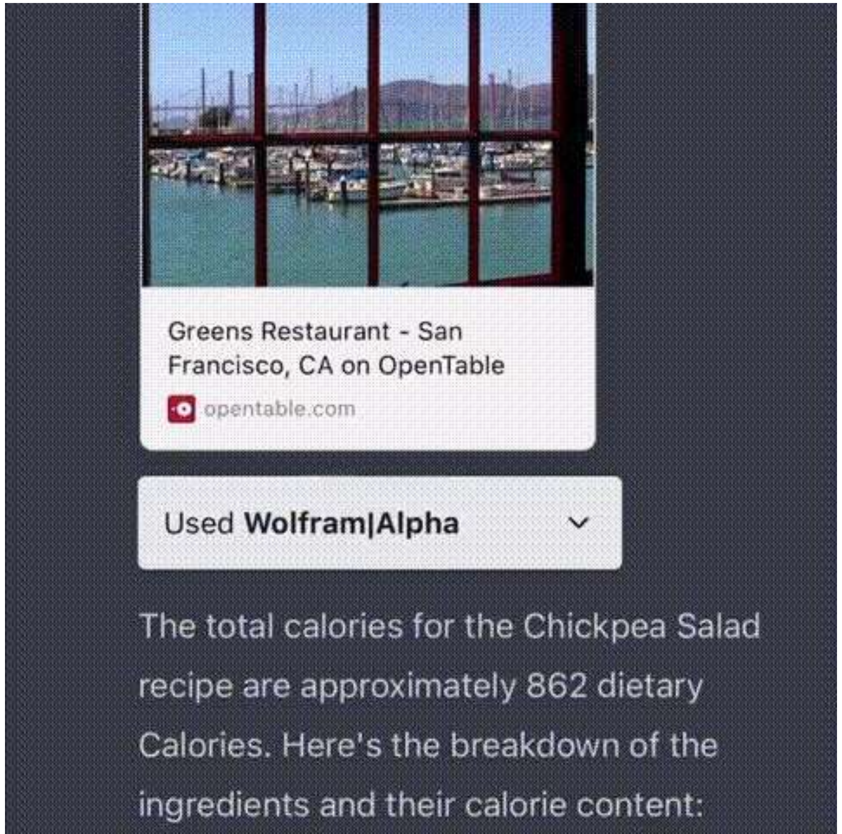
Project Background

The model not only needs access to external tools but must also be capable of independently planning and executing tasks. A generative AI model with reasoning, logic, and access to external information embodies the concept of an Agent. In other words, an Agent is a program that extends the out-of-the-box capabilities of a generative AI model.

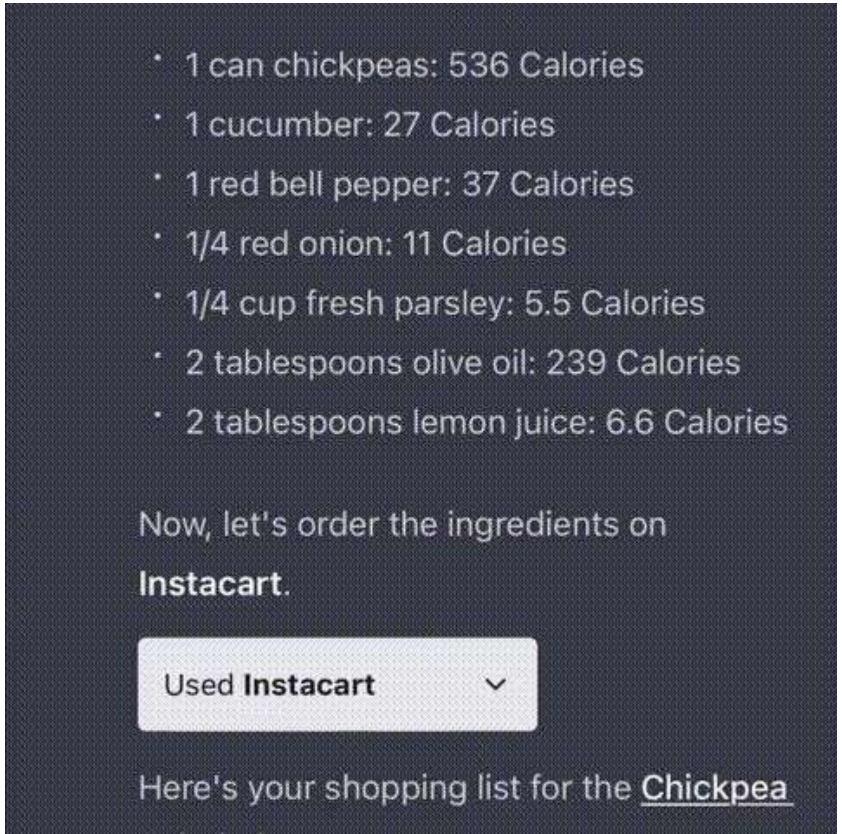
Find restaurants



Calculate calories



Order ingredients

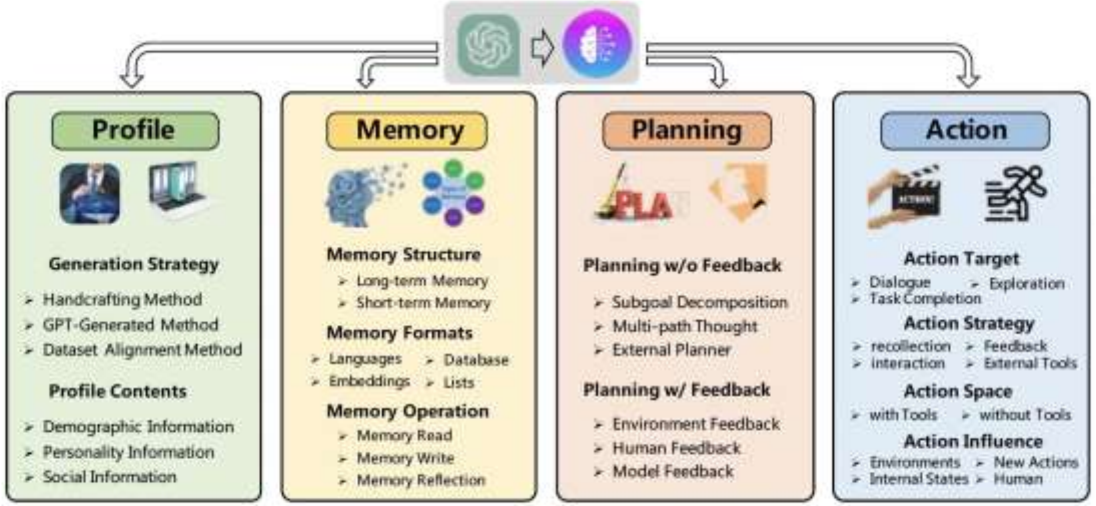


Project Background

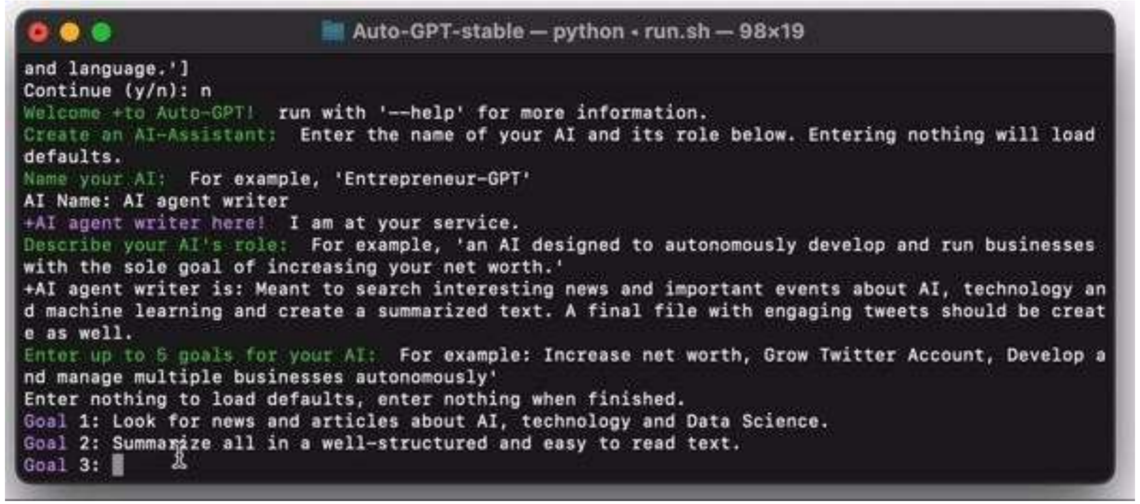
April 2023: Autonomous Agents

"An autonomous agent is a system situated within and a part of an environment that senses that environment and acts on it, over time, in pursuit of its own agenda and so as to effect what it senses in the future."

Franklin and Graesser (1997)



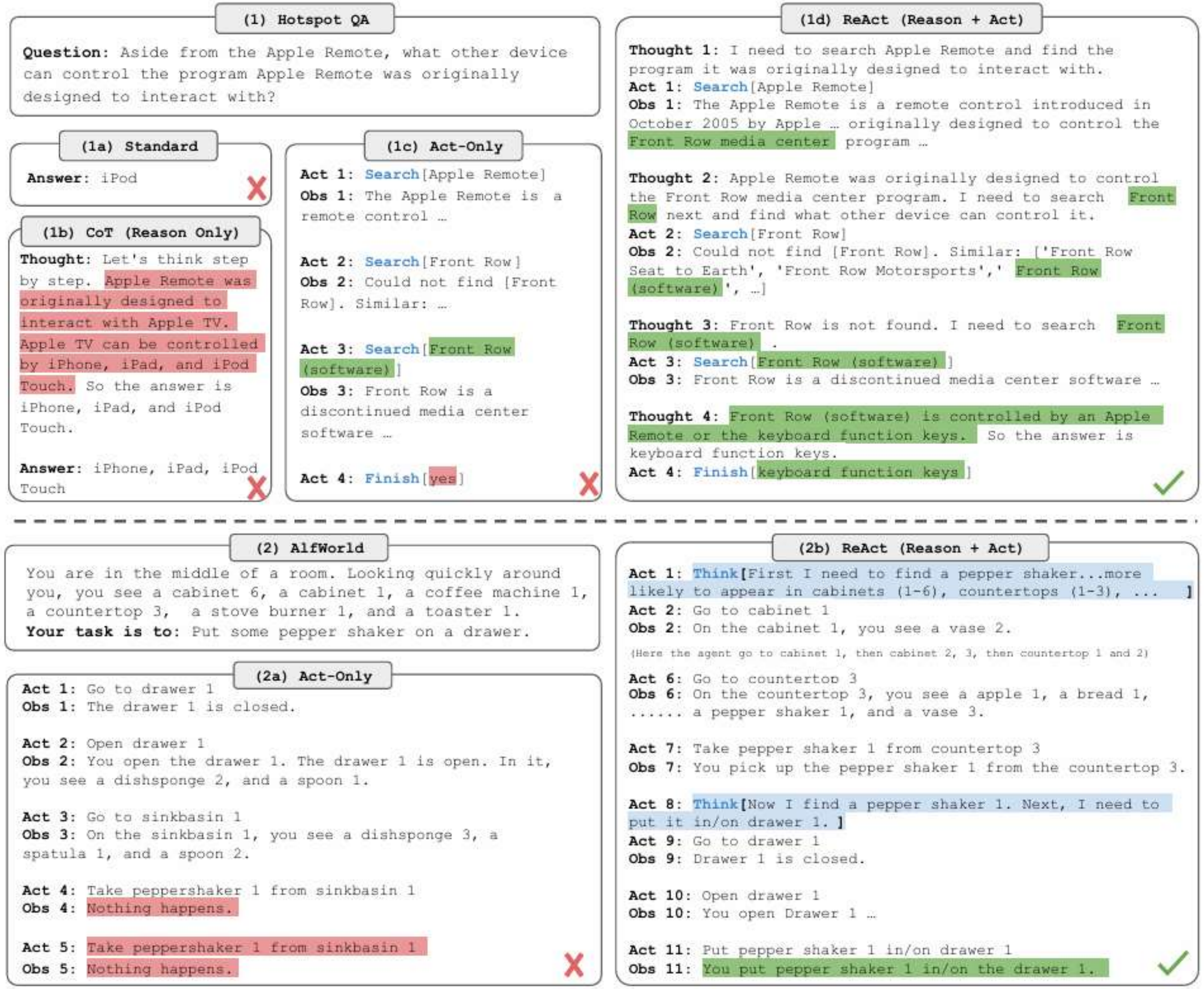
https://arxiv.org/abs/2308.11432
A Survey on Large Language Model based Autonomous Agents



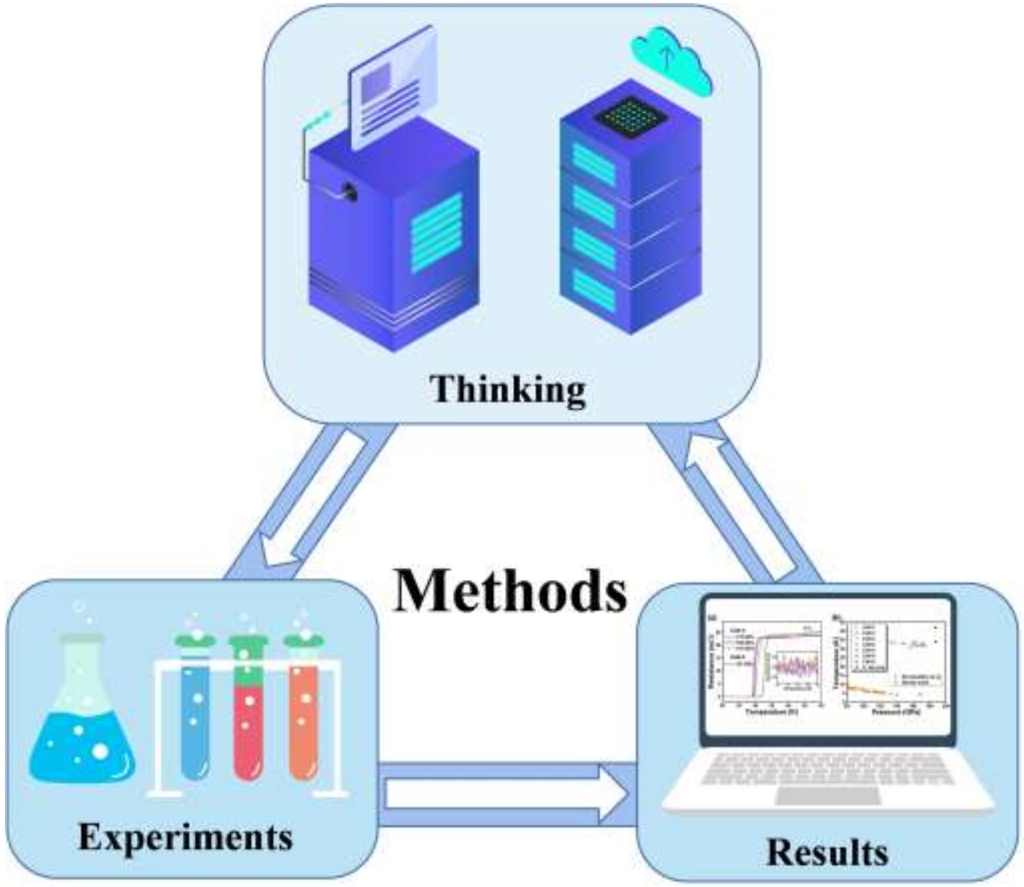
This is a pre-computed replay of a simulation that accompanies the paper entitled "Generative Agents: Interactive Simulacra of Human Behavior." It is for demonstration purposes only.

Project Background

ReAct: The Synergy of Reasoning and Action in Language Models

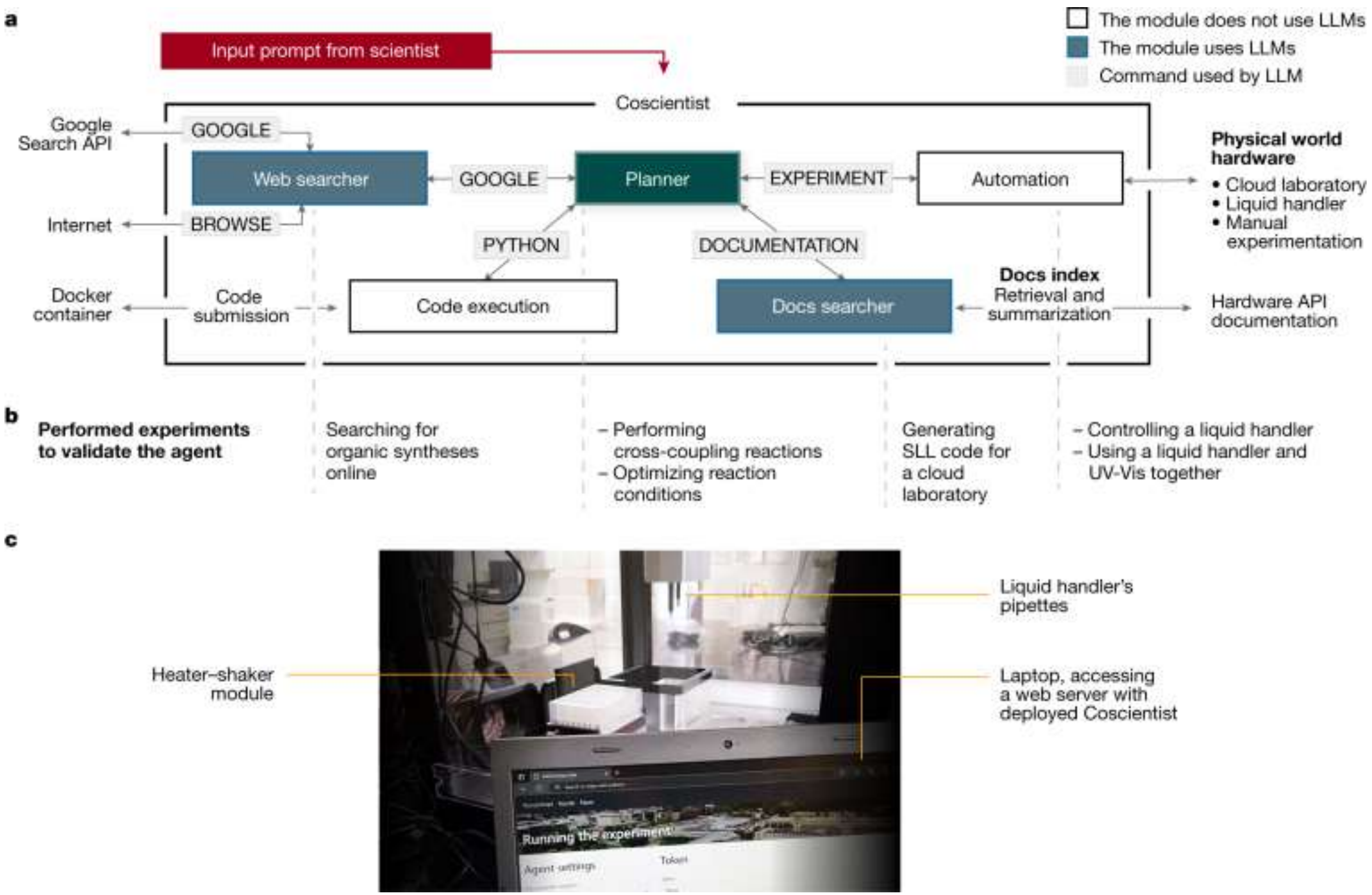


Experimental Disciplines



Project Background

AI Meets Chemistry: Turning the "Impossible" into the "Possible"



1.Online synthesis route planning

2.Search and organize literature

3.Cloud laboratory instructions

4.Precise control of reagent addition

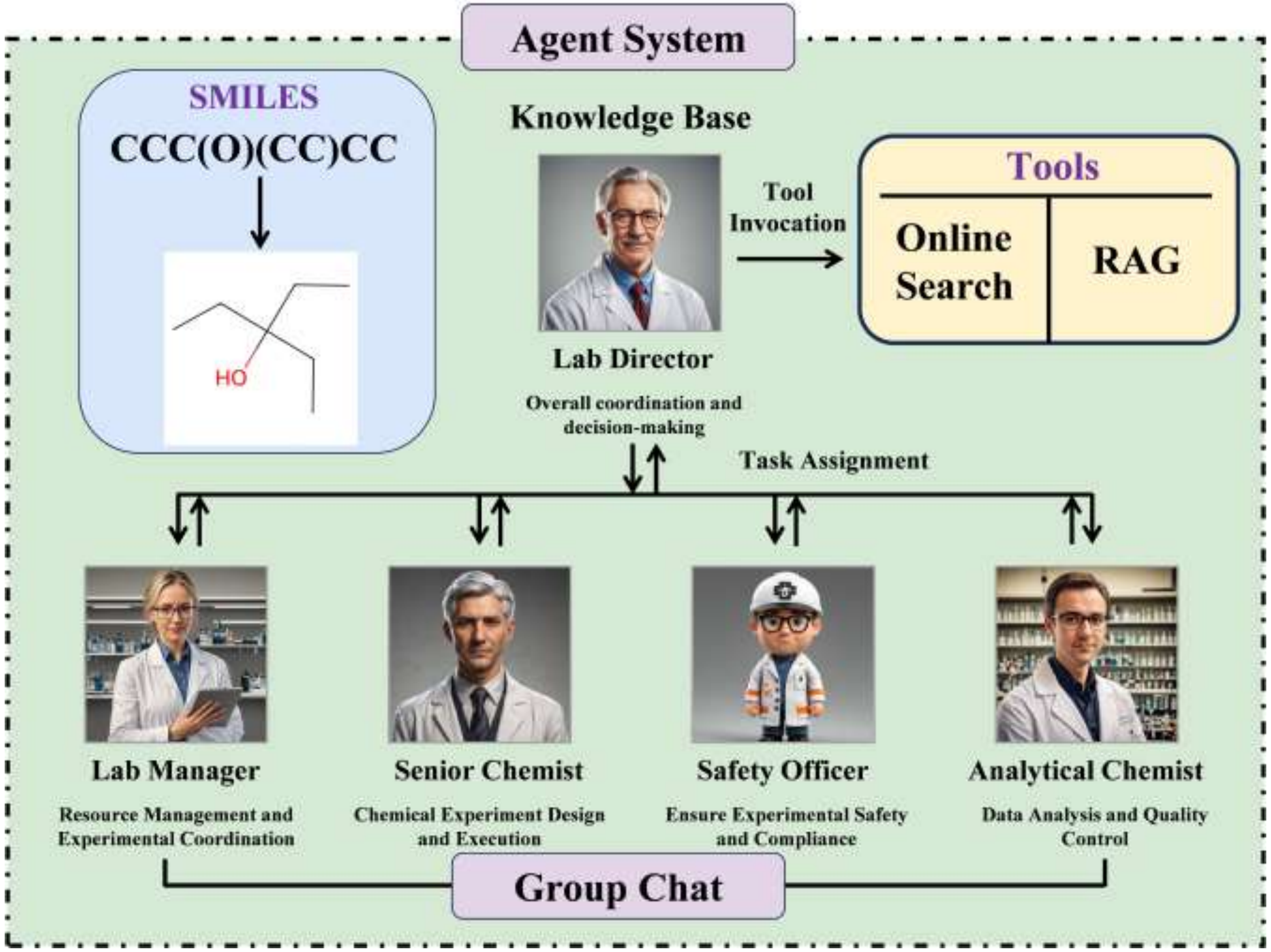
5.Complete complex synthesis

6.Result analysis and optimization

Basic Functions of GVIM

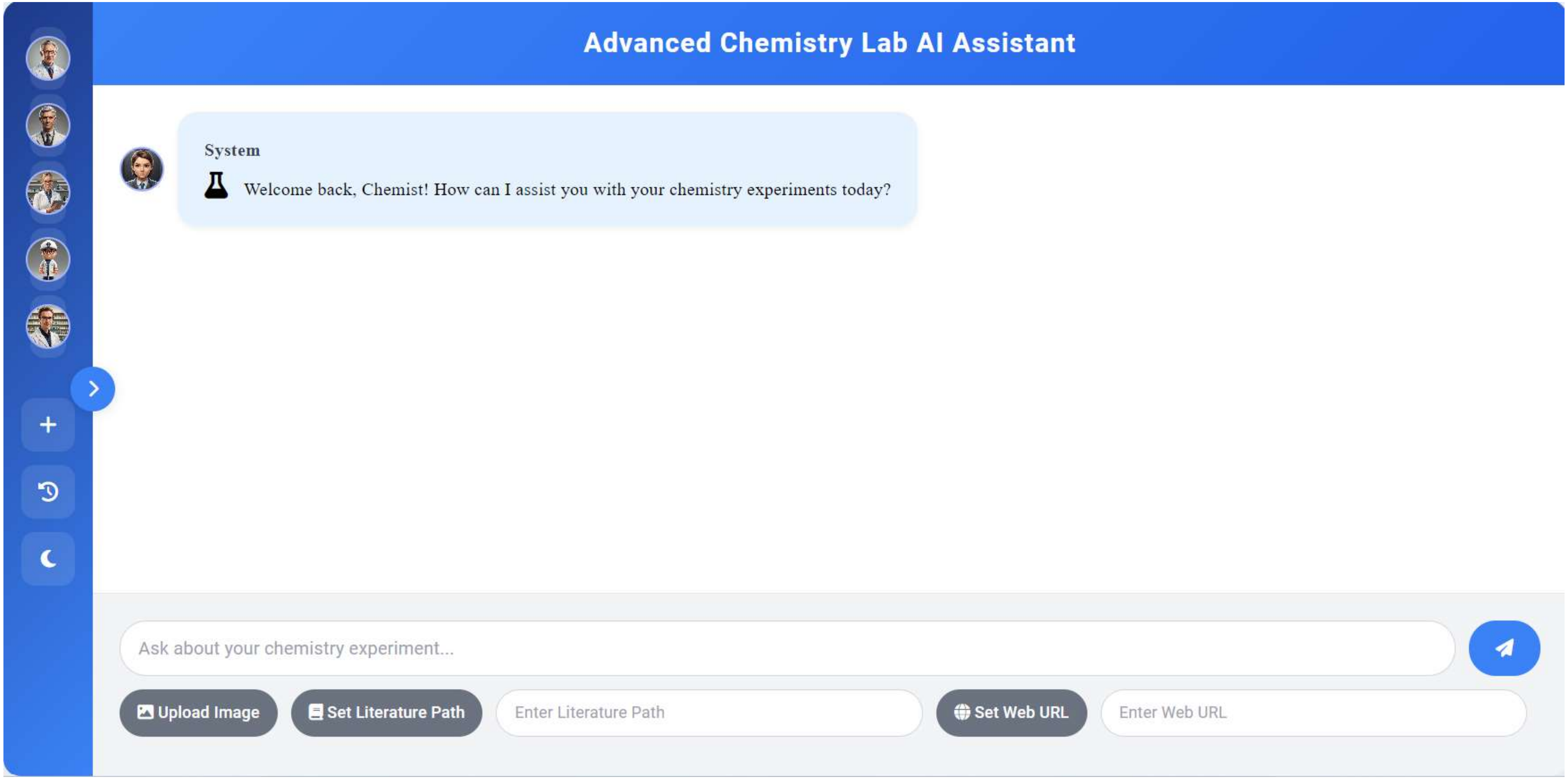
Basic Functions

Multi-Agent Collaboration



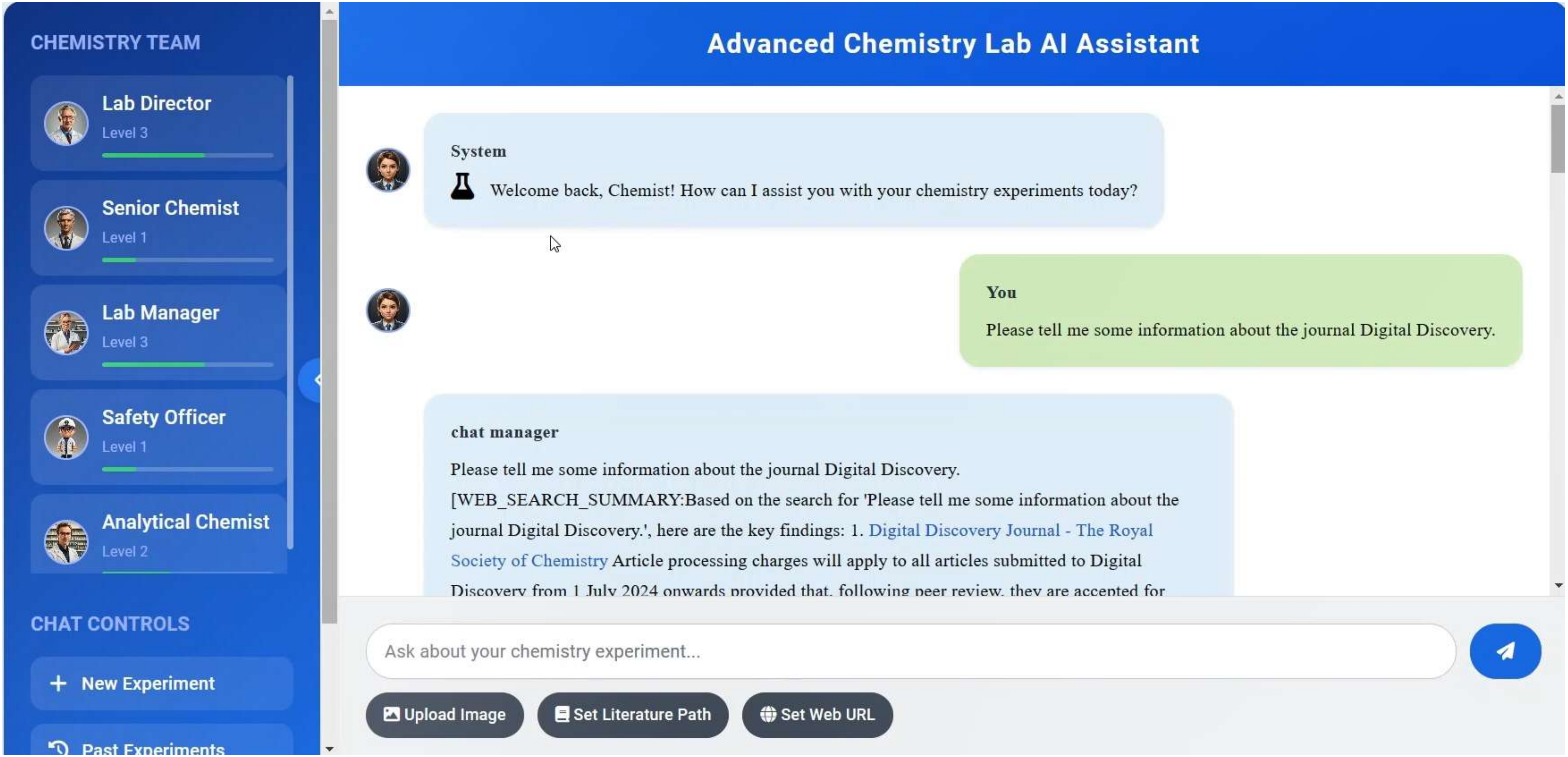
- **Multi-agent collaboration** simulates real human chemistry teams, providing multiple perspectives on chemical problems and proposing comprehensive and feasible solutions.
- **Online search** and **local knowledge base (RAG)** functionality (supports specified URL search and local document uploads).
- Use of **multimodal large models** to recognize handwritten chemical formulas.
- Visualization of SMILES characters in **3D and 2D structures.**
- AI-controlled browser functionality.
- Flexible use of **fine-tuned** or **non-fine-tuned** models.

Basic Functions



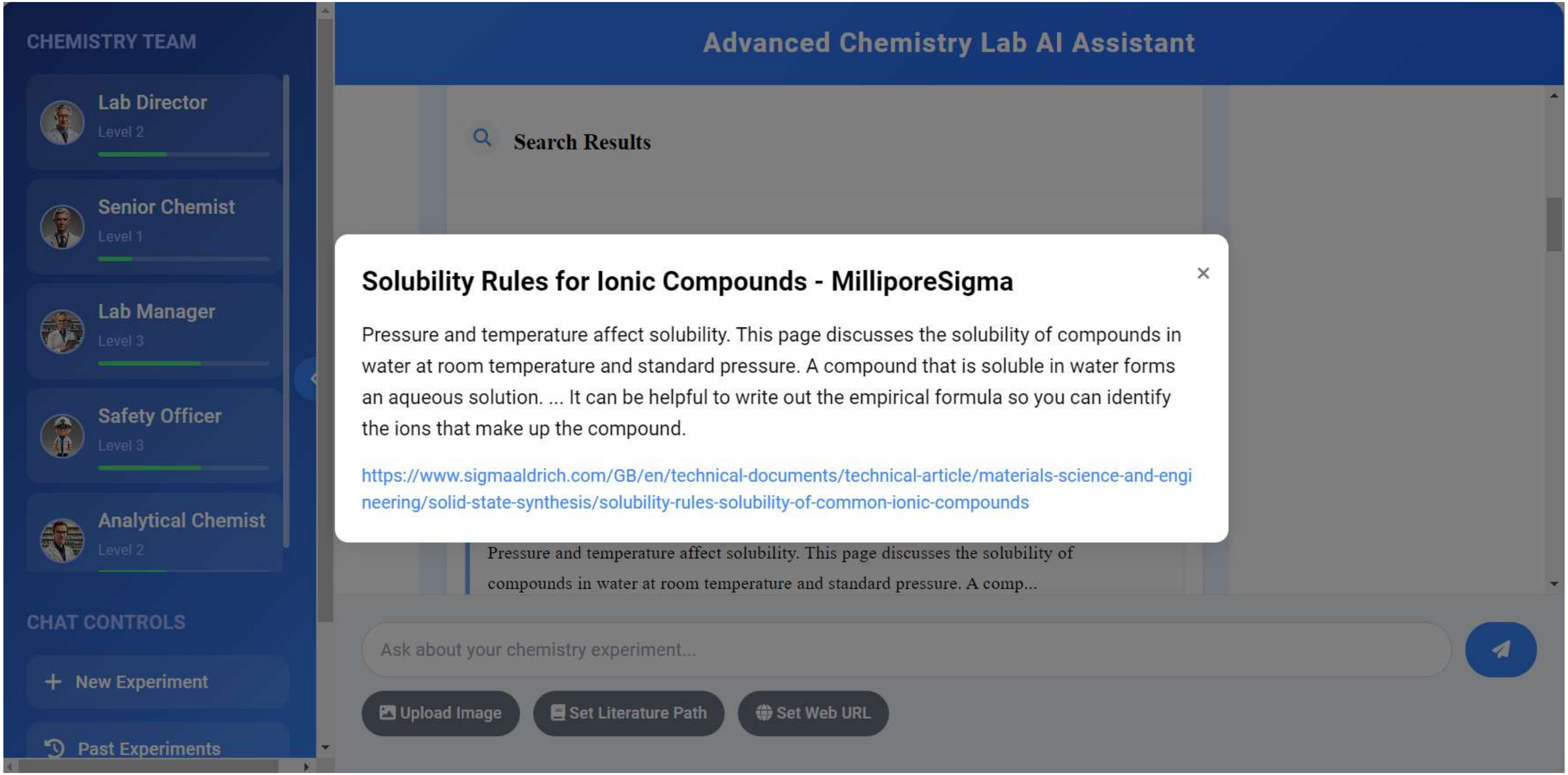
Basic Operation Interface

Basic Functions



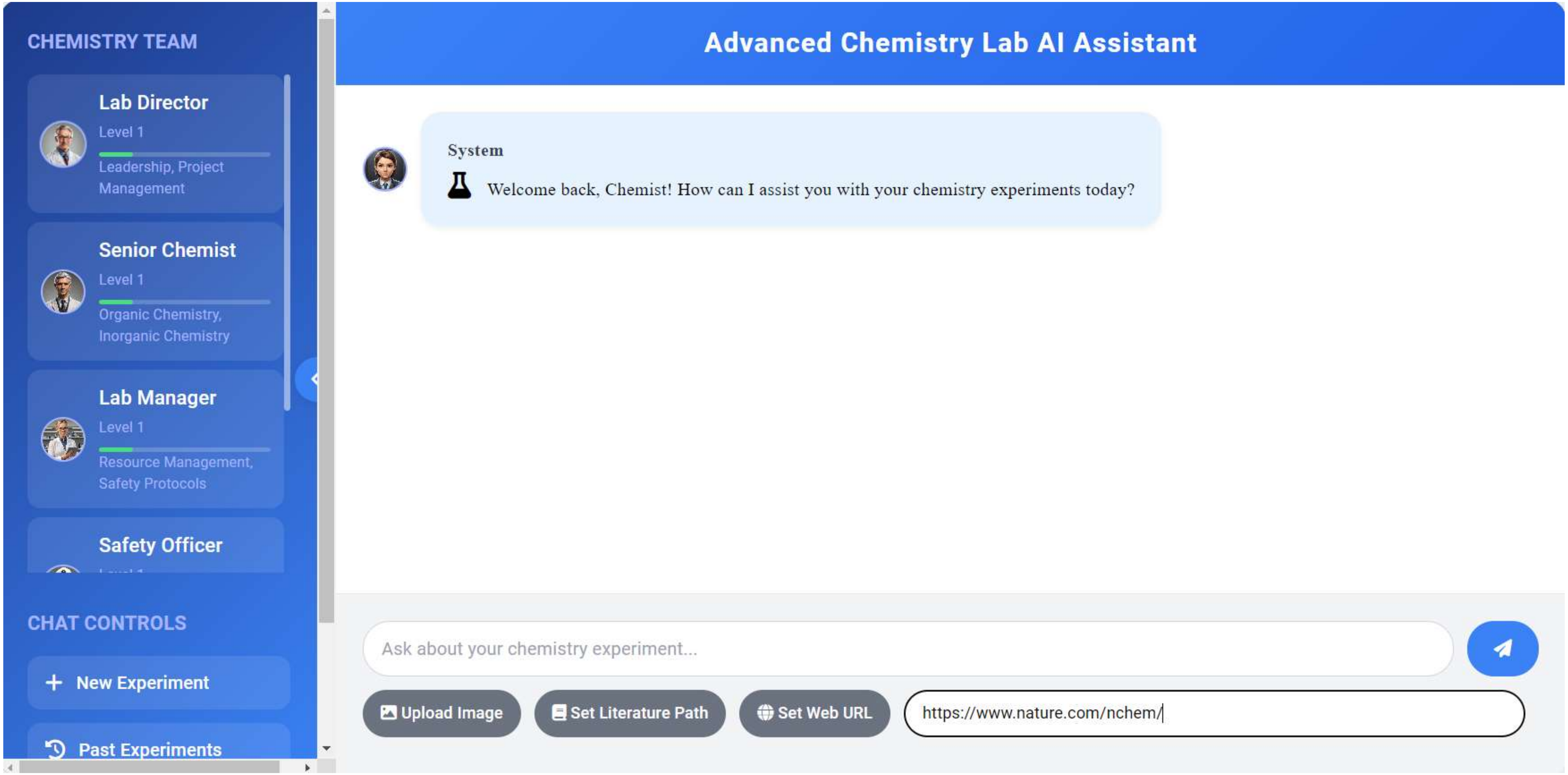
Online Search Function

Basic Functions



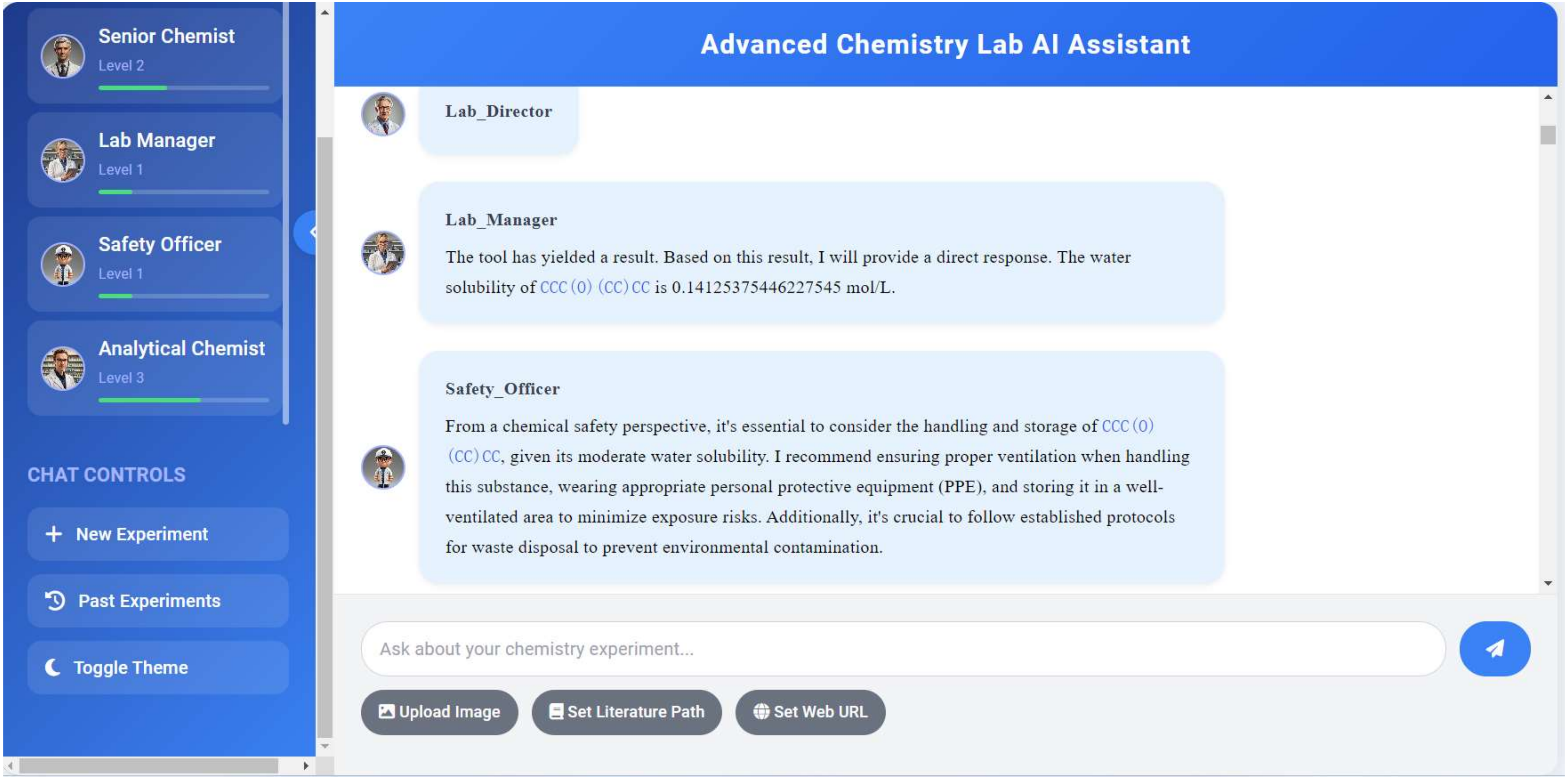
Online Search Result Preview

Basic Functions



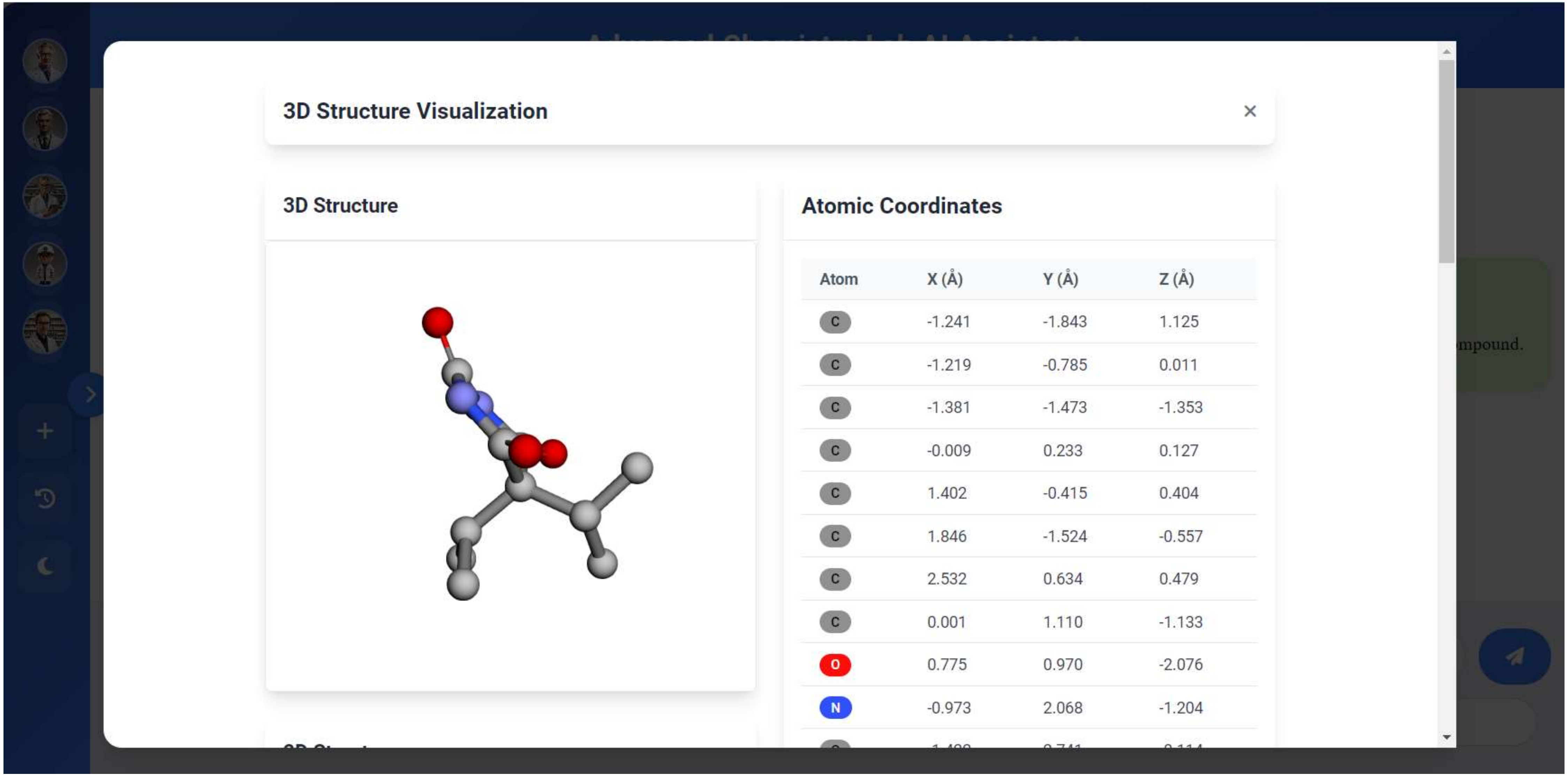
Specified URL Search

Basic Functions



RAG Based on Local Knowledge Base

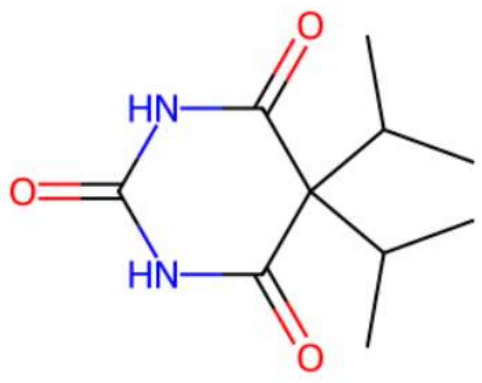
Basic Functions



SMILES Character 3D Structure Visualization

Basic Functions

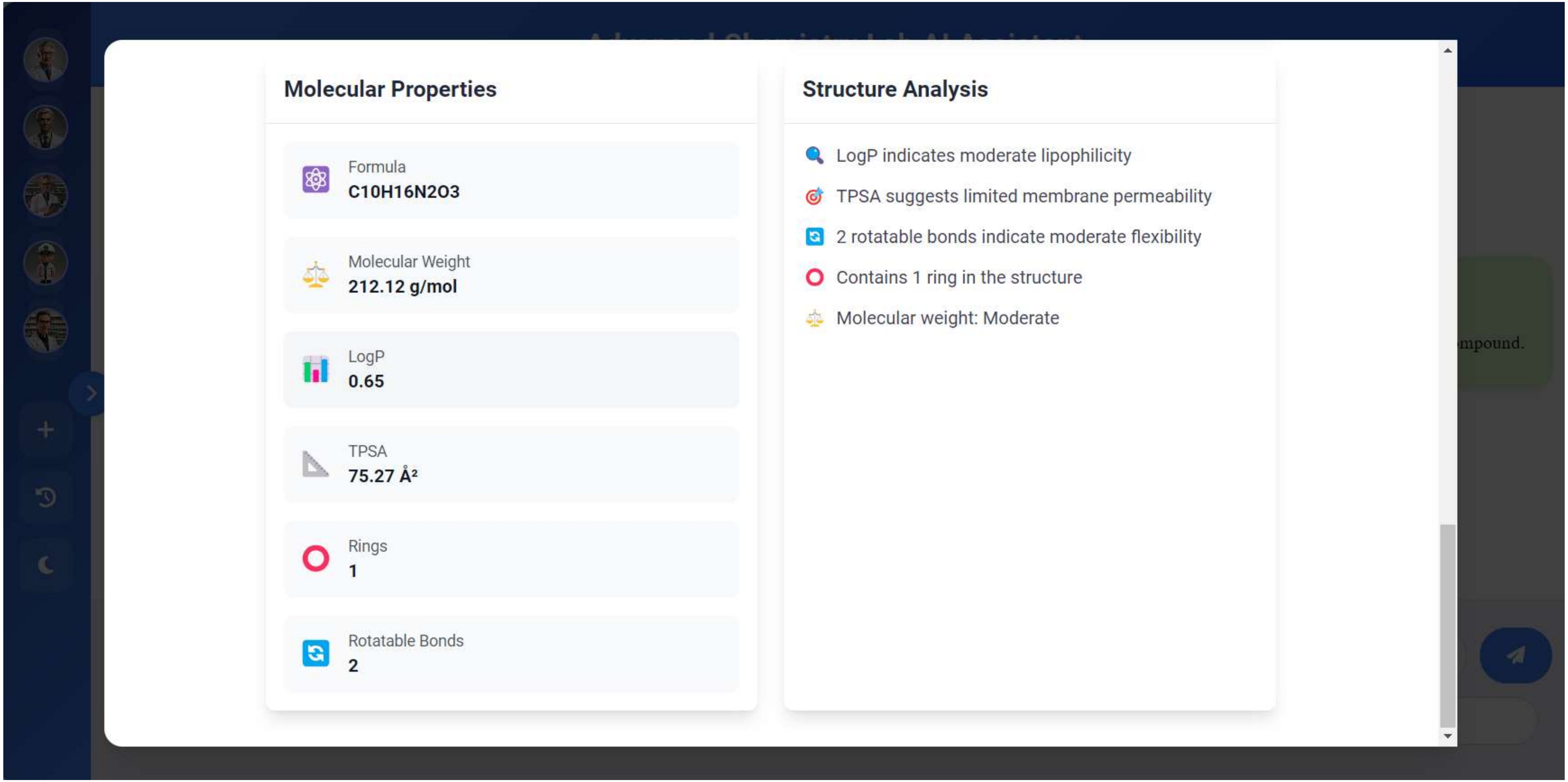
2D Structure



C	-1.422	2.741	-0.114
O	-1.862	3.882	-0.203
N	-1.272	2.131	1.090
C	-0.365	1.136	1.327
O	0.073	0.977	2.464
H	-2.168	-2.425	1.078
H	-1.201	-1.383	2.116
H	-0.412	-2.550	1.043
H	-2.149	-0.212	0.141
H	-1.521	-0.746	-2.159
H	-2.273	-2.111	-1.350
H	-0.535	-2.109	-1.614
H	1.367	-0.879	1.399
H	1.818	-1.208	-1.603
H	1.235	-2.422	-0.448
H	2.875	-1.832	-0.339

SMILES Character 2D Structure Visualization

Basic Functions



Introduction to Molecular Properties

Basic Functions

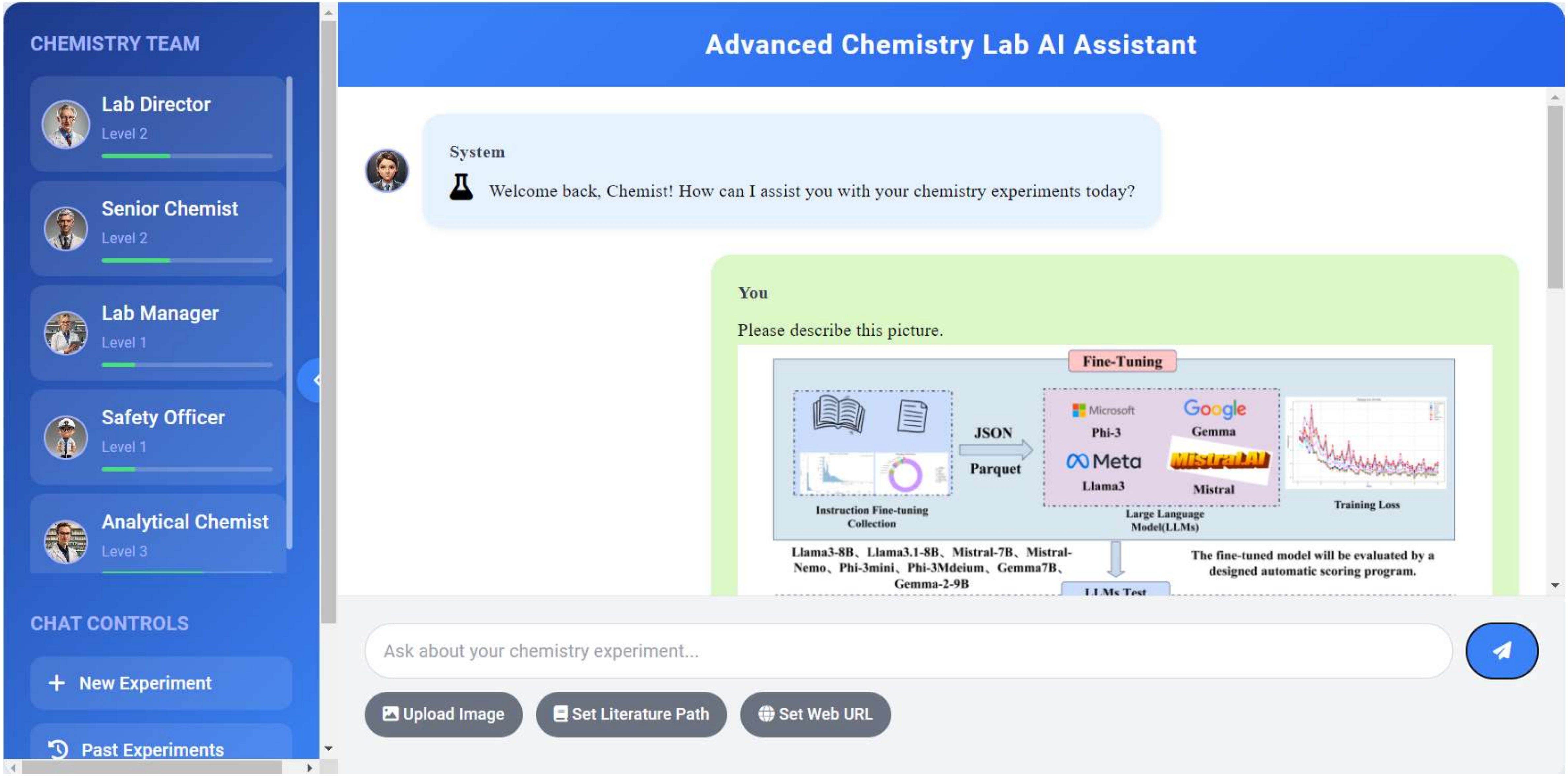
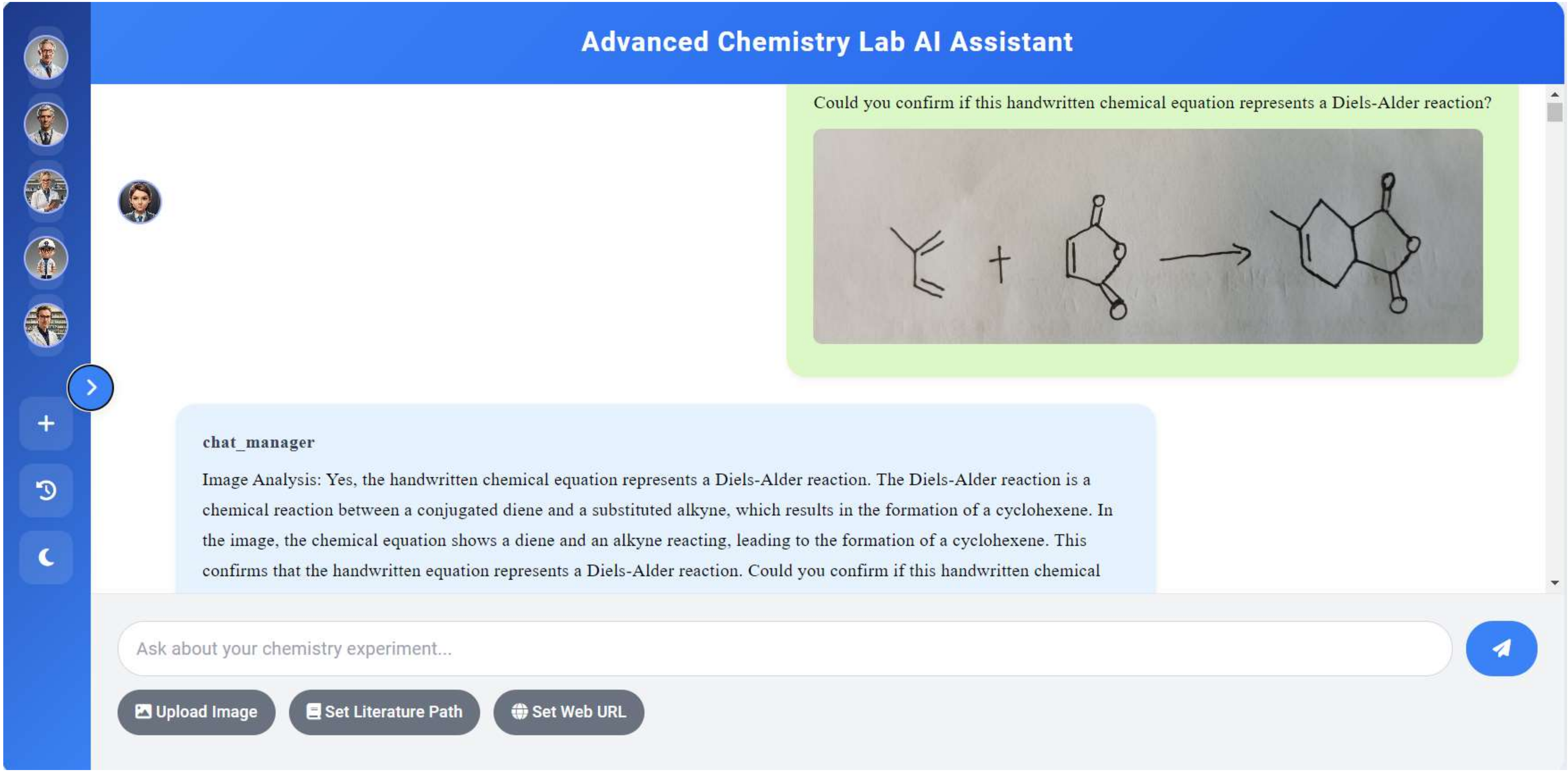


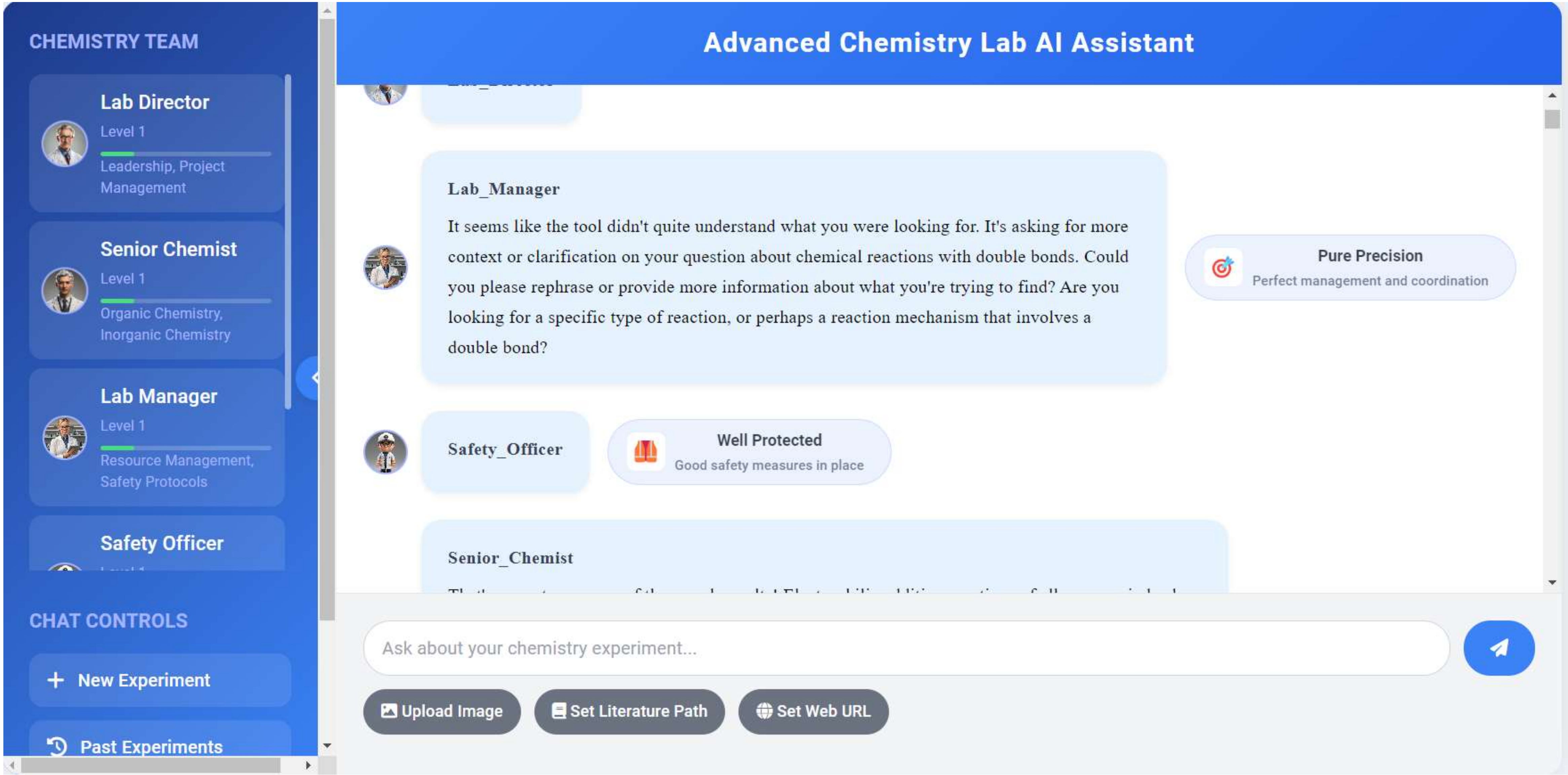
Image Recognition Description

Basic Functions



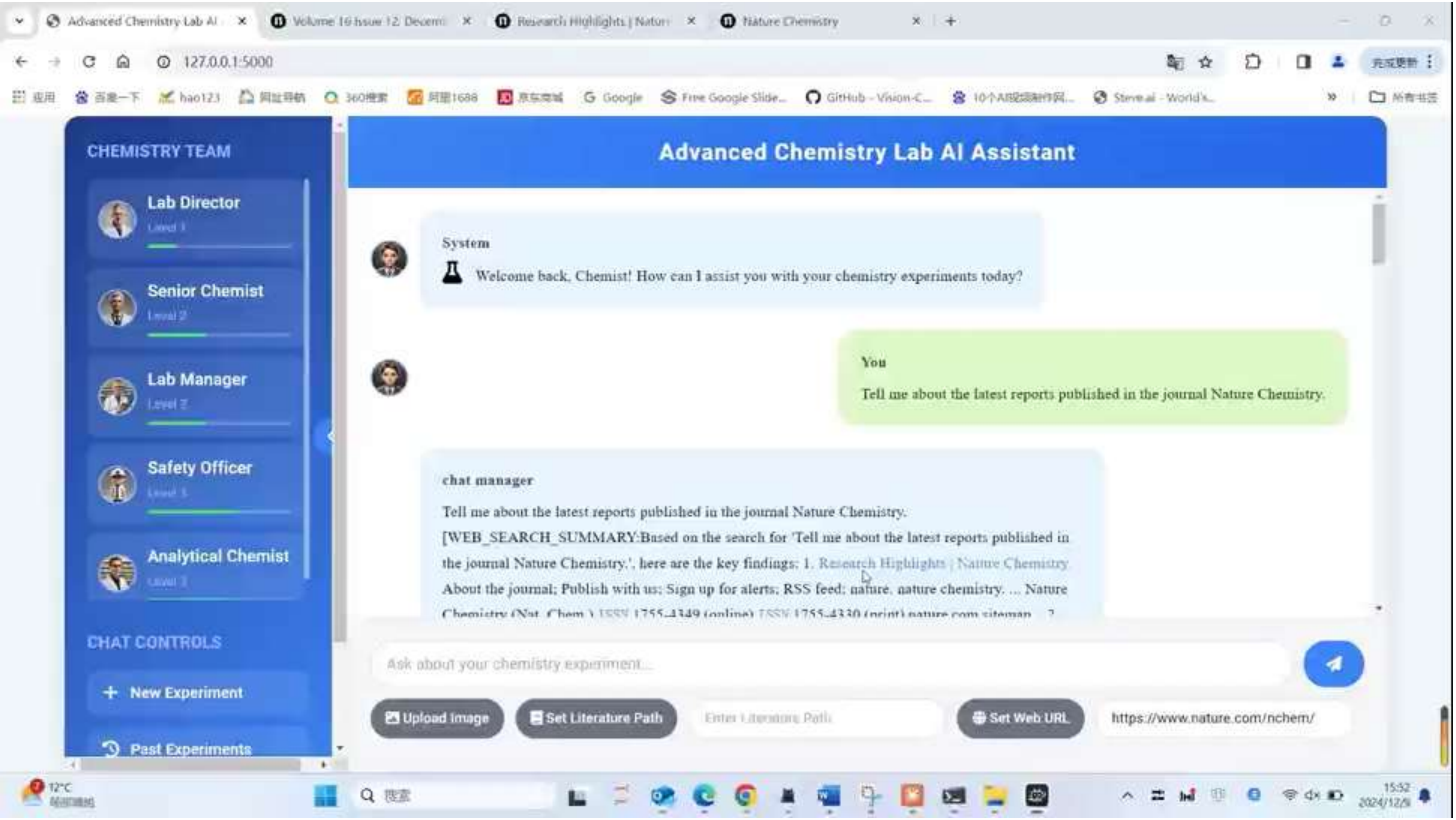
Handwritten Chemical Structure Recognition

Basic Functions



User Feedback

Demo of GVIM



Advanced Chemistry Lab AI Assistant
CHEMISTRY TEAM
Lab Director
Senior Chemist
Lab Manager
Safety Officer
Analytical Chemist
CHAT CONTROLS
New Experiment
Past Experiments
System
Welcome back, Chemist! How can I assist you with your chemistry experiments today?
You
Tell me about the latest reports published in the journal Nature Chemistry.
chat manager
Tell me about the latest reports published in the journal Nature Chemistry.
[WEB_SEARCH_SUMMARY:Based on the search for 'Tell me about the latest reports published in the journal Nature Chemistry.', here are the key findings: 1. Research Highlights | Nature Chemistry
About the journal; Publish with us; Sign up for alerts; RSS feed; nature. nature chemistry. ... Nature
Ask about your chemistry experiment...
Upload Image
Set Literature Path
Enter Literature Path
Set Web URL
https://www.nature.com/nchem/

Preliminary Results

Showcasing research from Kangyong Ma's laboratory, College of Physics and Electronic Information Engineering, Zhejiang Normal University, Zhejiang, China.

AI agents in chemical research: GVIM – an intelligent research assistant system

GVIM is a multi-agent-based chemical research system that operates through the collaborative work of professional role agents such as laboratory directors and senior chemists. This research also fine-tunes large language models using over 1.7 million collected chemistry domain instruction data points, enabling the system to provide professional chemistry model selection. GVIM can flexibly utilize different large language models, and combined with retrieval-augmented generation and chemical structure visualization capabilities, it provides new development ideas for chemical research.

As featured in:

See Kangyong Ma, *Digital Discovery*, 2025, 4, 355.

ROYAL SOCIETY OF CHEMISTRY

rsc.li/digitaldiscovery

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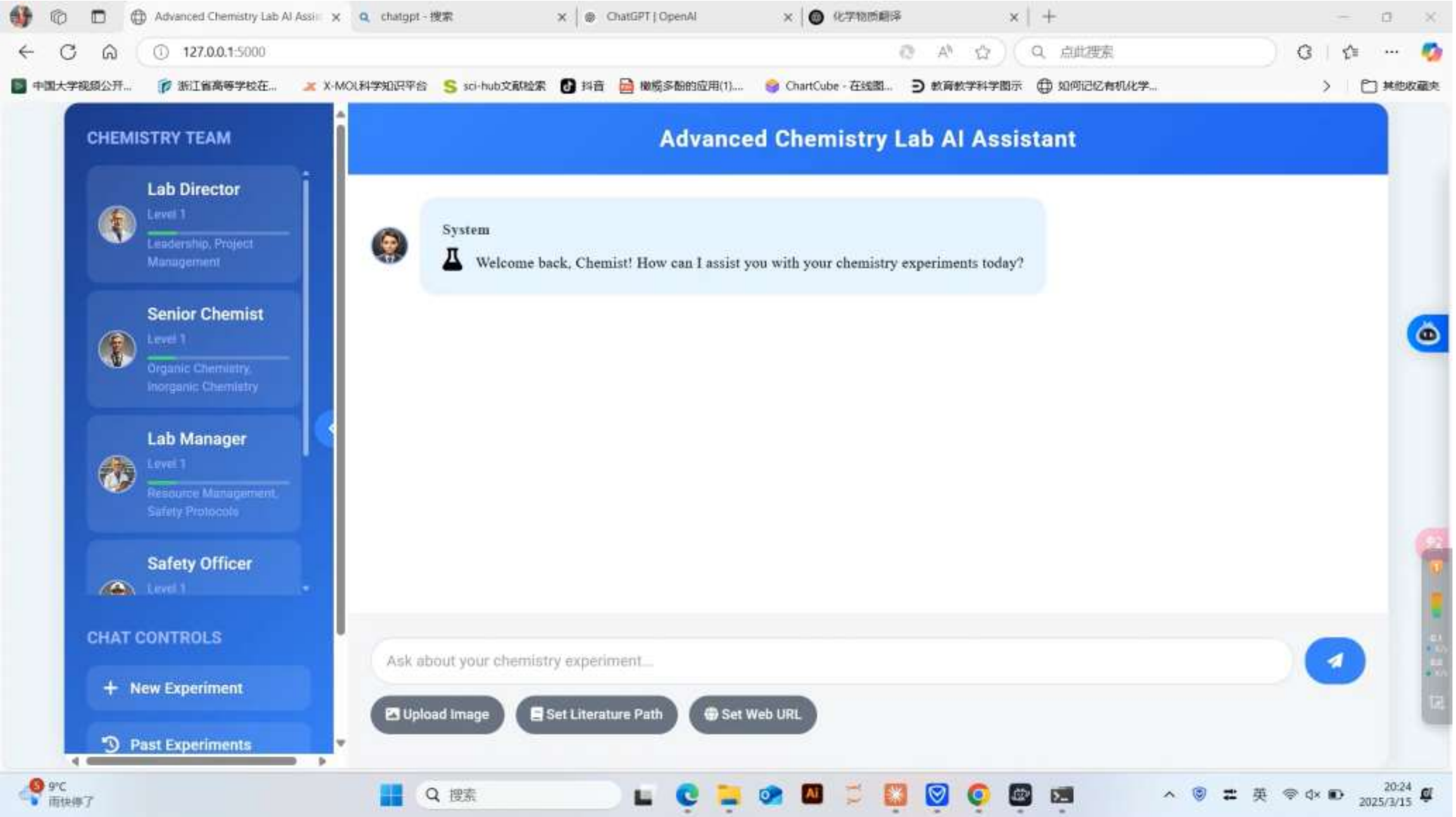
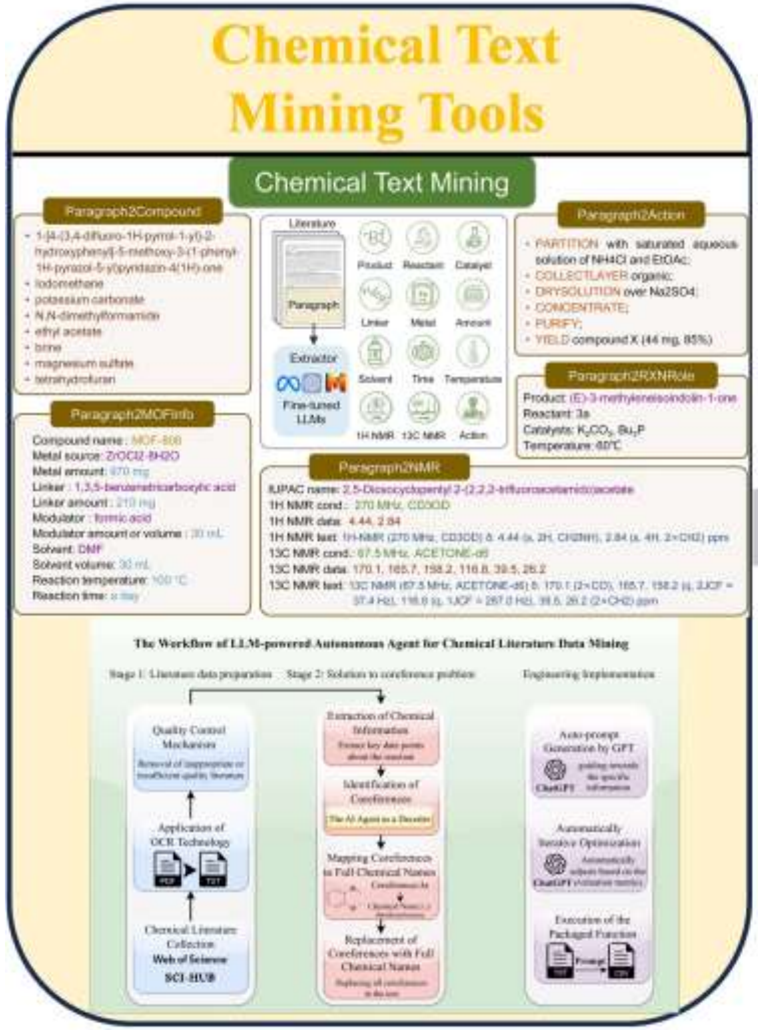
2025年02月25日

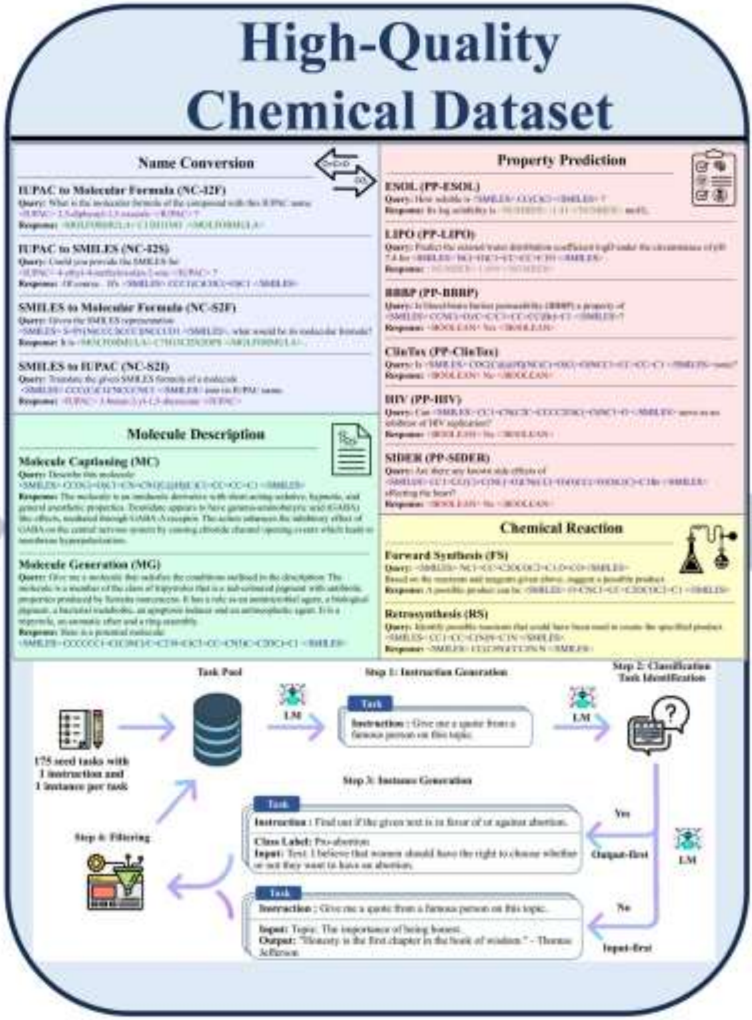
Future Prospects

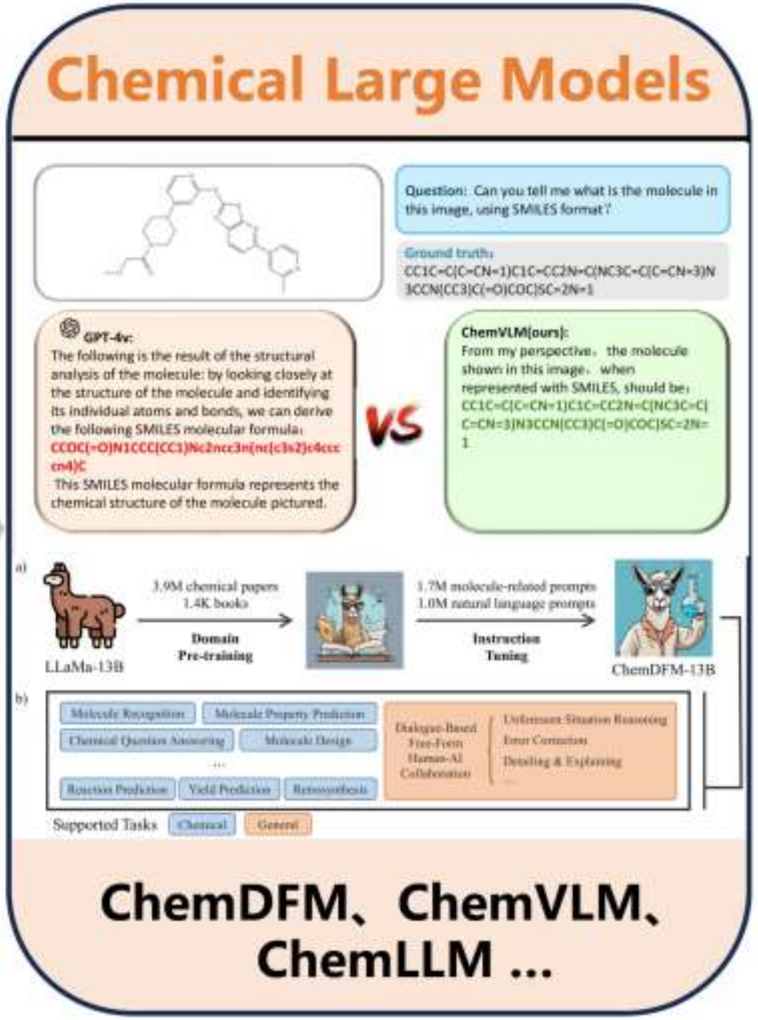


AI-Controlled Browser Functionality

Future Prospects

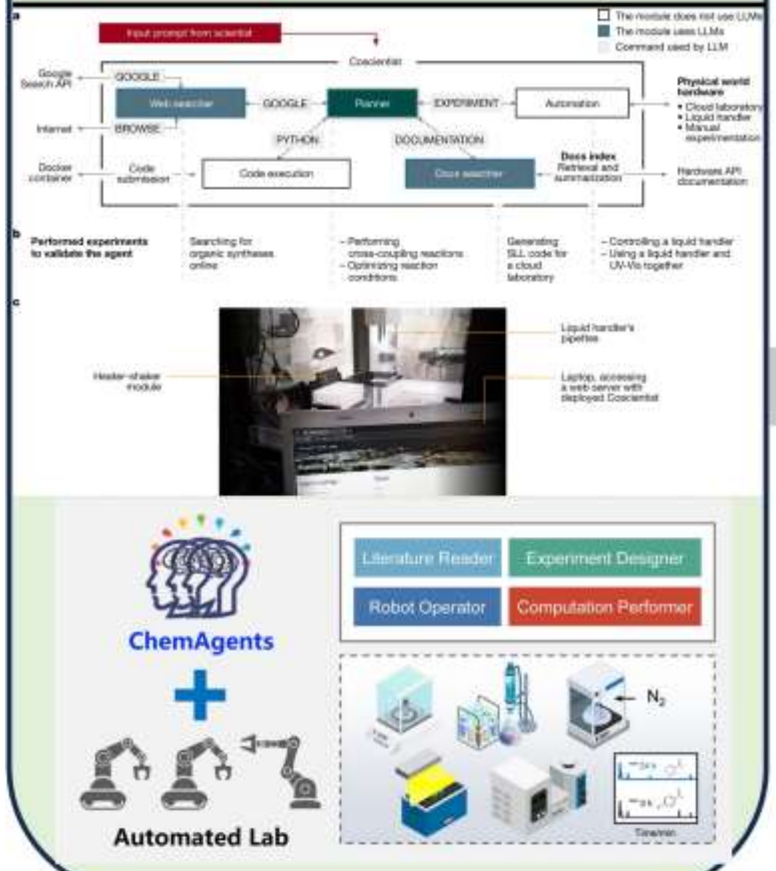






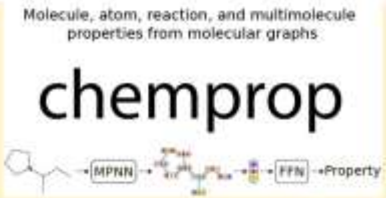
Future Prospects

Chemical Agents



MCP

PubChem



Software、Databases、Prediction Models ...

Embodied Intelligence

Figure 01、Unitree H1、Atlas...

Future Prospects

Chemical Research

Comprehensive Automation
High Intelligence